Benjamin B. McMillan

CONTACT INFORMATION

Institute for Basic Science, Center for Complex Geometry, B259
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RESEARCH INTERESTS

Cartan's method of equivalence, exterior differential systems, conservation laws, geometric structure of differential equations, invariant differential operators, singular foliations, quasiconformal geometry.

EDUCATION

UC Berkeley, Berkeley, CA

Ph.D., Mathematics, August 2010–May 2016

- Dissertation: "Geometry and Conservation Laws for a Class of Second-Order Parabolic Equations"
- Advisor: Robert L. Bryant
- National Science Foundation Graduate Research Fellowship, 2011

California Institute of Technology, Pasadena, CA

B.S. with honors, Mathematics, August 2006 – May 2010

ACADEMIC EXPERIENCE

Senior Researcher August 2022 – Present
Institute for Basic Science, Center for Complex Geometry, Daejeon, South Korea

Session Organizer, with Paul-Andi Nagy October 2025
Special Session "Symmetry and Integrability of Geometric Structures", 2025 KMS Annual Meeting

Referee
New Zealand Journal of Mathematics

Reviewer
American Mathematical Society, Mathematical Reviews

Australian Research Council Grant Funded Researcher September 2019 – May 2022
University of Adelaide, Adelaide, South Australia

Co-organizer, with Thomas Leistner August 2021 – May 2022
Differential Geometry Seminar

Course Coordinator and Lecturer March 2021 – June 2021
Analysis and Topology

RTG Postdoctoral Fellow August 2016 – June 2019
Stony Brook University, Stony Brook, New York, USA

Founding Organizer August 2017 – December 2017
Exterior Differential Systems and Method of Equivalence Seminar

Course Coordinator and Lecturer August 2016 – June 2019
Overview of Calculus, Differential Geometry of Curves and Surfaces, Ordinary Differential Equations, Introduction to Linear Algebra, Honors Linear Algebra

Lecturer August 2018 – December 2018
Calculus I

REU Mentor May 2017 – July 2017
Ricky Simanjuntak, calibrated planes in the Lie algebra of $SU(n)$ and the minimal cycles they determine.

Graduate Student August 2010 – May 2016
UC Berkeley, Berkeley, California, USA

Course Coordinator and Lecturer June 2011 – August 2011
Summer Semester Multivariable Calculus

Founding Organizer, with Christian Hilaire September 2012 – December 2013
Student Differential Geometry Seminar

Teaching Assistant August 2010 – May 2016
Pre-calculus, Calculus, Multivariable Calculus, Introductory Analysis, Discrete Math

Mentorship
UC Berkeley's directed reading program, guided undergraduates in self study of the topics:
- Galois Theory August 2014 – December 2014
- Classification of complex semi-simple Lie algebras January 2015 – May 2015
- Algebraic Topology January 2016 – May 2016

Tutor January 2012 – December 2012
UC Berkeley's New Experiences for Research & Diversity in Science

Undergraduate Student August 2006 – May 2010
California Institute of Technology, Pasadena, CA

Undergraduate Research
Research in Stochastic Evolution Equations June 2009 – August 2009
Research in Graph Theory June 2007 – August 2007

Tutor August 2009 – May 2010
Undergraduate Peers

Publications

L. MacDonald, B. McMillan, Chern–Weil theory for singular Haefliger foliations, *Differential Geom. Appl.*, to appear, (arXiv:2106.10078).

F. Costanza, M. Eastwood, T. Leistner, B. McMillan, A Calabi operator for Riemannian locally symmetric spaces, *Amer. J. Math.*, 148 (4) (2026), 1003–1028.

F. Costanza, M. Eastwood, T. Leistner, B. McMillan, The range of a connection and a Calabi operator for Lorentzian locally symmetric spaces, *Asian J. Math.*, 29 (2) (2025), 171–200.

B. McMillan, Geometry and conservation laws for a class of second-order parabolic equations II: conservation laws, *SIGMA Symmetry Integrability Geom. Methods Appl.*, 17:Paper No. 047, 24, 2021.

B. McMillan, Geometry and conservation laws for a class of second-order parabolic equations I: geometry, *J. Geom. Phys.*, 157:103824, 29, 2020.

C. Y. Ku and B. McMillan, Independent sets of maximal size in tensor powers of vertex transitive graphs, *J. Graph Theory*, 60 (4) (2009), 295–301.

Submitted:

B. McMillan, Indecomposable quasiconformal maps of manifolds, arXiv:2605.21894.

B. McMillan, Foliation surgeries and the concordance groups of foliated spheres, arXiv:2511.09053.

B. McMillan, A characterization of Pfaffian embeddings from (2,3,5)- into flat (4,7)-geometries, arXiv:2404.14988.

Selected Talks

Foliation surgeries and Godbillon–Vey spheres, 1/21/2026
Topology Seminar, Purdue University (online), invited

Foliation surgery and continuous variation of characteristic classes on spheres, 10/23/2025
Special Session, Symmetry and Integrability of Geometric Structures, KMS Annual Meeting

Secondary characteristic classes and Chern–Weil theory of singular foliations, 5/20–22/2025
Three-lecture series, Complex Geometry Seminar, IBS CCG, Daejeon

Invariant prolongation of the conformal Killing equation via conformal tractors, 4/25/2025
Geometry Session, KMS Spring Meeting

Embeddable (2,3,5) distributions, 10/18/2023
Workshop on Parabolic Geometry and Related Topics, IBS CCG, Daejeon, invited

The image of the Killing operator, 9/15/2022
Complex Geometry Seminar, IBS CCG, Daejeon

Calabi operators for locally symmetric spaces, 1/28/2022
QMAP Seminar, UC Davis (online), invited

The secondary characteristic classes of singular foliations, 12/8/2021
Special Session, Geometry with Symmetries, AustMS Annual Meeting

Conservation laws and parabolic Monge–Ampère equations, 1/17/2020
Special Session, Geometry of Differential Equations, AMS JMM, invited

Conservation laws and parabolic Monge–Ampère equations, 12/5/2019
Special Session, Geometry including Differential Geometry, AustMS Annual Meeting

Conservation laws and the moduli space of solutions to PDE, 3/7/2019
Topology Seminar, Ohio State University, invited

Conservation laws and parabolic Monge–Ampère equations, 9/25/2018
Geometry/Topology Seminar, Stony Brook University

The geometry and conservation laws of parabolic equations, 10/25/2016
Topology Seminar, Ohio State University, invited

The geometry and conservation laws of parabolic equations, 9/27/2016
Geometric Analysis seminar, Rutgers, invited

Moving frames and geometric invariants, 4/5/2014
Graduate Student Topology and Geometry Conference, UT Austin

The Newlander–Nirenberg theorem, 2/27/2014
The Cartan Seminar, Stanford